# Need for Speed: Hydrodynamic Performance

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***Abstract*— A robot was constructed to compare the hydrodynamics of cone and dome shaped nose cones. The purpose of the experiment was to compare simulation to reality in a non-ideal environment to consider how both methods can be used to predict the behavior of underwater vehicles. The robot measured its own velocity along with the applied drag force to the nose cones. These values were then used to determine the associated coefficients of drag. The coefficient of drag is a dimensionless quantity used to describe the hydrodynamics of a shape. The wind velocity was also collected to account for offsets from the environment. Simulated coefficients of drag were obtained using COMSOL, a multiphysic software tool. A laminar flow simulation with models of the nose cones was used to relate force to flow velocity. The coefficient of drag for the cone-shaped nose was found to be 0.959±0.185 in experiment and 0.636±$10^{-4}$ in the simulation. The coefficient of drag for the dome-shaped nose was found to be 1.44±0.235 in the experiment and 0.696±$10^{-4}$ in the simulation. Thus, it can be said that the cone-shaped nose is more hydrodynamic than the dome-shaped nose.**

***Keywords*— Hydrodynamics, Nose Cone, Drag, Coefficient of Drag, Underwater Technology, COMSOL**

## I. Introduction

The goals of the experiment were to measure the drag force due to each nose cone at a known constant velocity to determine the coefficient of drag. The expected takeaways from the data were determining which shape would suit a hydrodynamic vessel to make it faster and more energy efficient and evaluating the method for accurate results [1]. The smaller the coefficient of drag is, the faster the terminal velocity will be at a certain thrust input. The quality of the method would be determined by its similarity to the simulated experiment. It is important to use both ideal simulations and physical experiments because physical experiments have more factors that can't be accounted for in a simulation's idealized environment.

The coefficient of drag is determined by drag force (D), fluid density (r), velocity (V), and the area of the projection of the object perpendicular to the direction of fluid flow (A) [2,3].

$$C_d = \frac{2D}{rAV^2} \tag{1}$$

### A. Engineering Goals

This project seeks to compare and investigate the hydrodynamics of differently shaped vessels, examining the relationship between simulation and in situ testing in the ocean.

## II. Sensor Selection

All sensors used in the design are connected to a Teensy 4.0. The measurement range for the Teensy is 0V to 3.3V. The Teensy measures the voltage coming into its analog pins at a rate of 10Hz [4]. The circuit board is powered by a 12.6V battery. A voltage regulator steps down the voltage to 5.043V which is used to power the circuit board.

### *A. Velocity Sensor*

A velocity sensor is necessary to obtain the value of velocity in Equation 1. This experiment calculates the velocity through the use of two pressure sensors to emulate a pitot tube. The pitot tube measures pressure from the incoming air which is replaced by a forward-facing pressure sensor ($p_t$), and the pressure at rest which is replaced by a downward-facing pressure sensor ($p_s$) [5]. The velocity of the fluid (V) is determined by the fluid density (r), and the pressure differential between the total and static pressure (Equation 2).

$$V^2 = \frac{2(p_t - p_s)}{r} \tag{2}$$

The pressure sensor chosen was an MPX5700. When the pressure sensor is powered with 5V, the expected output range is 0.810V to 0.839V.

### *B. Anemometer*

An anemometer was selected to measure wind speeds as the wind can impact the velocity of the robot. While aiming to investigate the hydrodynamics of different nose shapes below the surface, the robot's performance is impacted by more than just the shape of the nose. With a surface bot, wind can easily alter the robot's course and speed.

The anemometer was built using the reference design provided on the E80 website [6]. Three 3D-printed cups are mounted on an axle (Figure 1), thus the anemometer rotates when there is a significant amount of wind present, which wind tunnel testing revealed to be more than 1.8 m/s. This is due to the cup size of the anemometer and resistance due to friction of the axel and ball bearing.

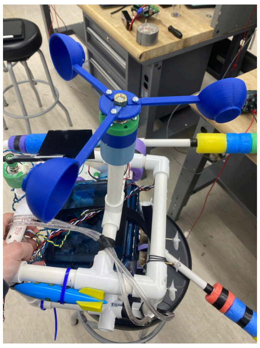

Fig. 1 Anemometer Mounted on Robot

To record the wind speeds, the AH9246 Hall effect sensor was employed. The Hall effect arises when an electrically charged material is in the presence of a magnet. Electrons move in a straight line, however, a magnet curves the path of electrons as it repels the magnetic field of the electrical current. From the displacement of electrons, a Hall voltage is measured by the Hall effect sensor [7]. Thus by implementing a magnet attached to the axle of the anemometer, a hall effect sensor is the ideal choice for tracking rotations. Surrounding the axel and aligned with the magnet is a housing for a hall effect sensor. When the magnet passes the hall effect sensor the voltage spikes. The frequency of these spikes reveals the frequency at which the anemometer is spinning. When powered with 5V, the hall effects sensor outputs 0mV or 70mV when in proximity to the magnet [8]. To convert from frequency of the anemometer to wind speed, a calibration curve was created using the wind tunnel (Figure 2). The wind speed was measured using the manometer while 30 seconds worth of rotations were counted at each wind speed. There is a linear relationship (Equation 3) between the frequency of the anemometer and wind speed.

$$V = 1.85 \frac{m}{rotation} \cdot f \frac{rotation}{s} + 1.8 \frac{m}{s} \quad (3)$$

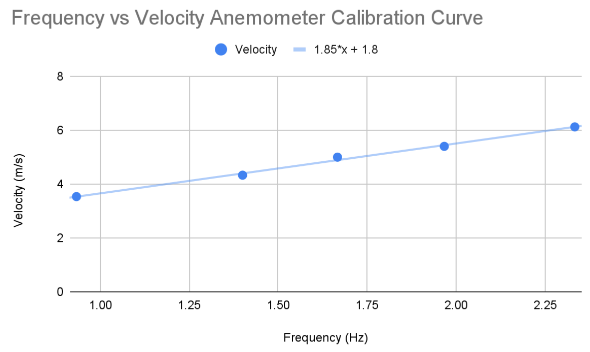


Fig. 2 Anemometer calibration curve

Since the frequency of the anemometer never exceeds 10 Hz the Teensy sampling rate is sufficient.

*C. Weather Vane*

The weathervane consists of a 3D printed fin, counterbalanced with three, 6-32 x ⅜" screws and nuts which was built in accordance with the E80's "Reference WeatherVane" document [9]. Marine epoxy was used to hold the fin upright as well to waterproof the wires. The gathered information operates in conjunction with the anemometer data to predict the path the robot may veer off to. The weather vane is mounted on a radially rotating resistor that allows it to convey wind direction information. A potentiometer was suitable for measuring wind direction because it was easy to implement and purely mechanical so was adverse to failure. This voltage reading is then mapped to the orientation in degrees that the weathervane is pointing. The relationship between voltage output and wind direction is shown in Figure 3.

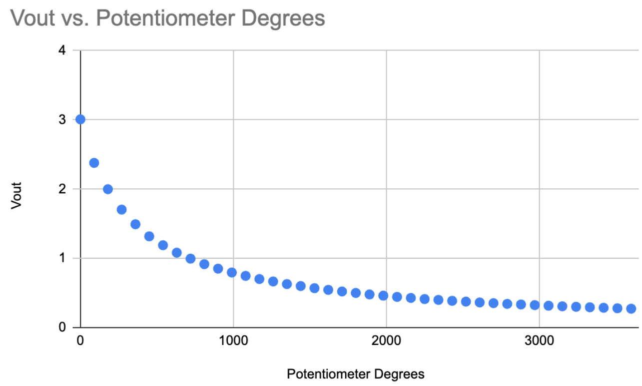


Fig. 3 Weather vane calibration curve

*D. Force Sensor*

Fig. 4 Force sensor housing

To measure the performance of the nose cones underwater a force sensor (Figure 4) was placed behind the nose cone. The force sensor measures the effective drag force experienced by the nose cone when the robot is in motion. When the sensor is powered by 5V the expected output is 1.8V to 3.0V. The 1.8V is the voltage read by the teensy when the robot is not in motion. This is because the housing and the water pressure produce an unaccounted-for force on the pressure sensor. The maximum expected output voltage is 3V. The maximum force is calculated using Equation 1 and the expected velocity of the robot.

## III. Modeling

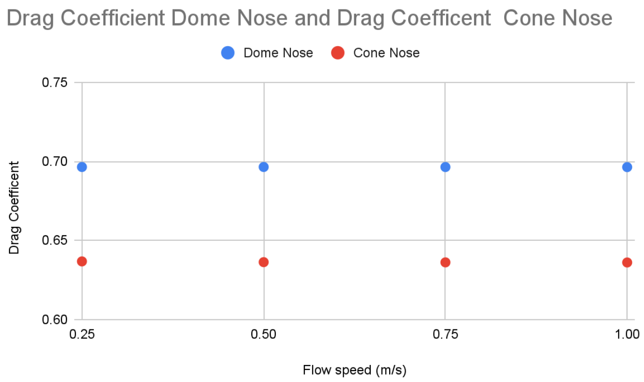


Fig. 5 Simulated coefficient of drag for the dome-shaped nose and cone nose across flow speeds

A Laminar flow simulation was done in COMSOL to obtain the relationship between drag force and flow velocity for each nose cone. The coefficient of drag is expected to be constant under any flow speed so the resulting values were averaged over to obtain the theoretical coefficient of drag for each nose cone (Figure 5). The averaged theoretical coefficient of drag for the dome nose was 0.696 and for the cone nose 0.636.

## IV. Circuit and Robot Design

### A. Mechanical Design

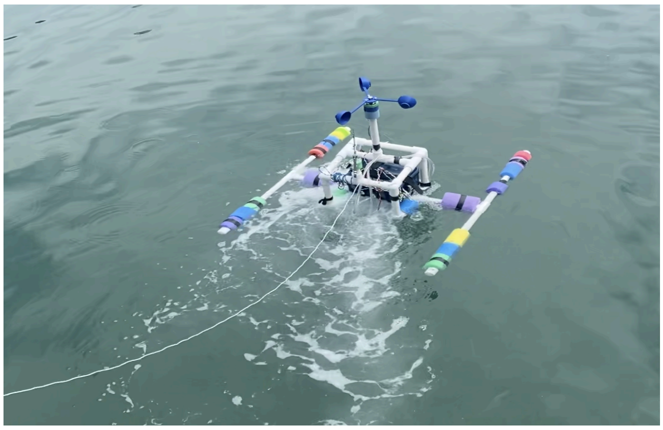

Fig. 6 Extending arms making the robot buoyant during deployment

The robot construction used PVC pipes as a frame to hold the waterproof box that housed the circuitry. All exposed wire junctions were waterproofed using heat shrink and parafilm. Behind the box are 3 horizontally aligned motors to propel the robot forward. Extending arms with parts of pool noodles were attached to the sides to stabilize the robot (Figure 6). An arm extending below the surface of the water holds the nose cones (Figure 7,8) and force sensor housing (Figure 9).

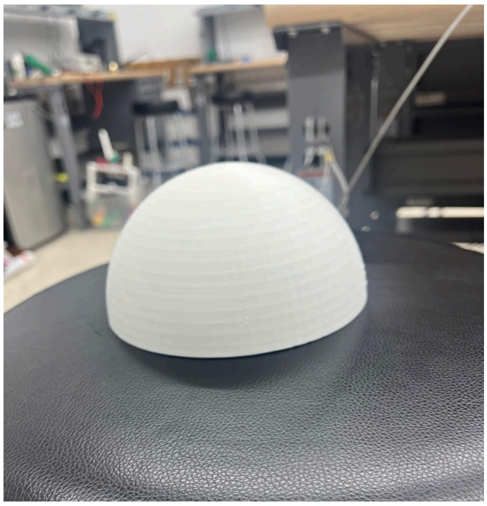

Fig. 7 Dome nose

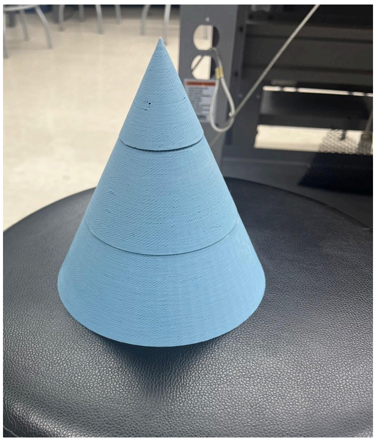

Fig. 8  Cone nose

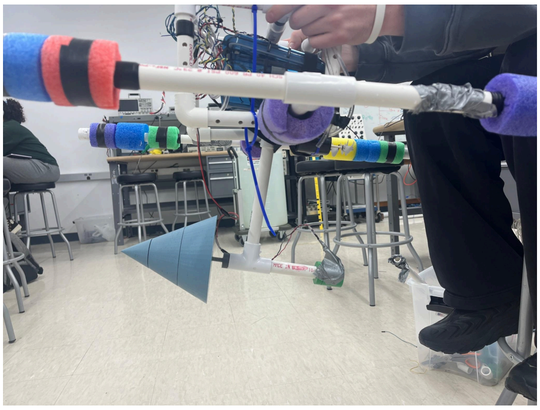

Fig. 9 Underwater portion of the robot depicted with the cone nose

The robot was designed as a surface craft so the nose cones could be held stable and dominate the drag force underwater. The nose cones each have a ¼” nut placed in the back which screws into a bolt attached to a piece of PVC. The bolt runs through the length of a PVC tube which at the other end holds the force sensor housing (Figure 10).

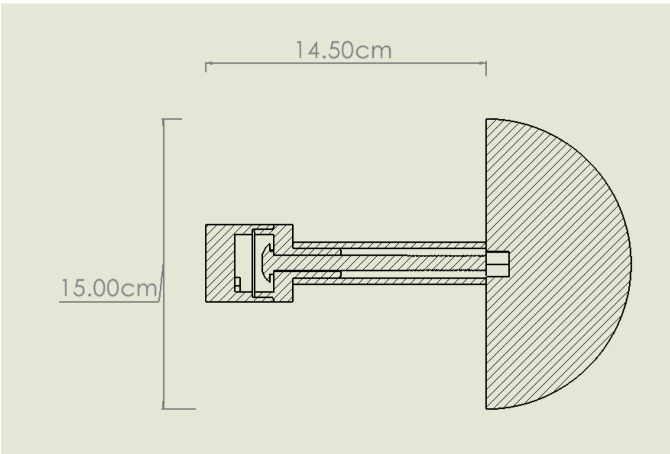


Fig. 10  Force sensor apparatus diagram

The force sensor housing (Figure 4) is a cylindrical container that has a parafilm barrier waterproofing the force sensor.

Fig. 11  Weathervane [9]

The end of the bolt pushes against the force sensor inducing a voltage reading. Housings for the wind direction and speed sensors are attached at the top to be away from the surface (Figure 11,12). The anemometer was attached to a PVC stand so that it would be well above the waterline and free of any water interference. In addition to being well above the waterline, the sensor was epoxied in place so that it was waterproofed. To attach the anemometer, a part was designed that screwed into the hall effect sensor housing that fits snugly in the PVC (Figure 12).

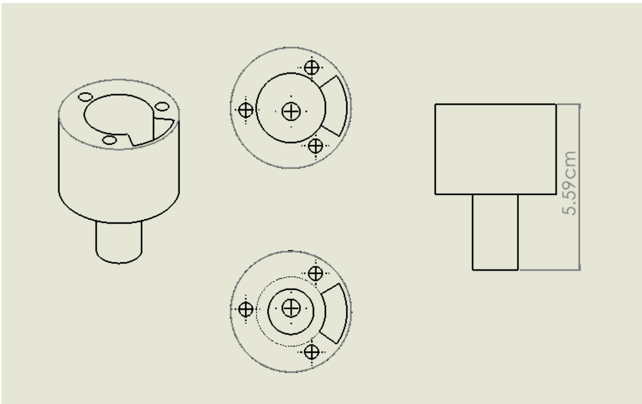


Fig. 12 PVC to Anemometer Adapter

The pressure sensors are connected to plastic hoses taped to the robot frame to mimic a pitot tube (Figure 13).

Fig. 13  Pressure tubes on robot frame

*B. Velocity Sensor*

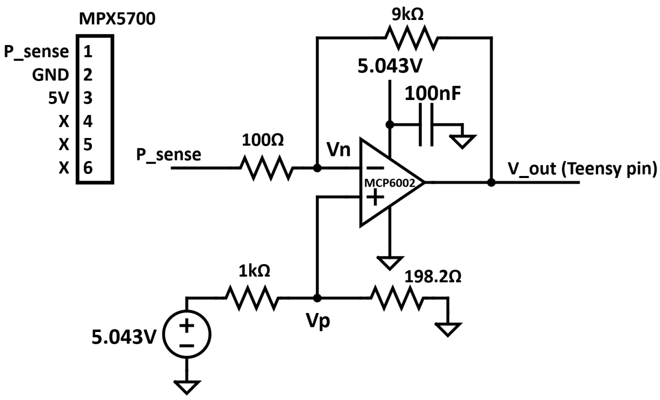


Fig. 14  Velocity sensor circuit schematic

The velocity sensor uses two identical MPX5700 pressure sensors acting as a pitot tube. The 0.810V input representing atmospheric pressure is mapped to a 3V output to the teensy and the 0.839V input representing the maximum expected pressure is mapped to a 0.3V output to the teensy. The maximum velocity measured by the output is 3.6 meters per second. This is greater than the maximum expected velocity, but it is implemented to ensure all data is within the voltage range accepted by the Teensy and any unexpected results are captured. The circuit was designed before additional mechanical modifications were added, so this buffer is overcompensating. Thus, the expected voltage range does not use the whole measurable range from the teensy. The motors were tested using a force measurement stand to determine that the expected total thrust force is approximately 300N when fully submerged. Equating this value to Equation 1 and solving for velocity results in an estimated maximum velocity of 0.8 m/s. The circuit used is an offset amplifier with a gain of 90 and an offset of ~76V. The ideal circuit should have an offset of ~70V but due to the sensitivity of the circuit, the battery input voltage of 5.043V instead of 5V meant the grounded resistor in the positive branch of the op-amp (Figure 14) needed to be changed from 200Ω to 198.2Ω resulting in an optimal offset of 76V. The output voltages are converted to pressure using the associated calibration curves and then to velocity using Equation 2. The resulting total pressure and static pressure calibration equations are shown below.

Total Pressure:

$$-22.114\frac{kPa}{V}\cdot V_{p_{sense}} + 69.524kPa \tag{4}$$

Static Pressure:

$$-22.087\frac{kPa}{V}\cdot V_{p_{sense}} + 66.266kPa \tag{5}$$

The teensy sample rate of 10 Hz should be sufficient for a pressure sensor because it is a DC output and unlikely to change significantly in 0.1 seconds. According to the MPX5700 datasheet, the response time is 1 millisecond which is sufficient for the experiment [10].

*C. Anemometer*

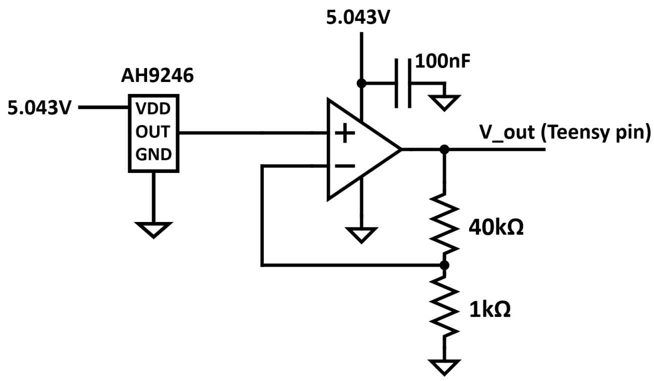


Fig. 15  Anemometer circuit schematic

In testing, the AH9246 hall effect sensor provided output voltages of 0V when in the proximity of a magnet and 70mV without a magnet under a 5.043V input voltage. To get the output voltage in the desired teensy range a non-inverting op amp circuit was used to amplify the sensor output (Figure 15). The circuit consisted of an MP601 op-amp and a 40kΩ and 1 kΩ resistor. The voltage divider created by the resistors amplified the voltage output as shown in Equation 6.

$$V_{out} = 0.07(\frac{40k\Omega}{1k\Omega} + 1) = 2.9V \tag{6}$$

*D. Weathervane*

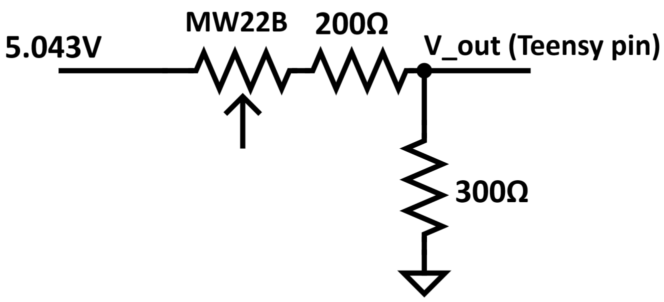


Fig. 16 Weathervane circuit schematic

The weathervane is affixed to an MW22B potentiometer, which has a resistance range between 0kΩ to 5kΩ, found through testing. This resistance range was selected to match the variation in output, ensuring compatibility with the 0V to 3.3V input range of the Teensy. As depicted in Figure 16, the circuit incorporates a series connection of a 200Ω and a 300Ω resistor. The voltage output ($V_{out}$) ranges between 3.008V to 0.2730V when the potentiometer is fully turned. Equation 7 provides the formula for calculating the voltage, which can then be mapped to the degree the weathervane is pointing:

$$V_{out} = 5.043V \cdot \frac{300\ \Omega}{MW22B + 500\ \Omega} \quad (7)$$

The circuit is powered by a 5.043V input source, which was chosen to provide sufficient power for the operation of the potentiometer. The sample rate of 10Hz was deemed sufficient for this application because it is unlikely for the wind direction to change significantly in 0.1sec.

*E. Force Sensor*

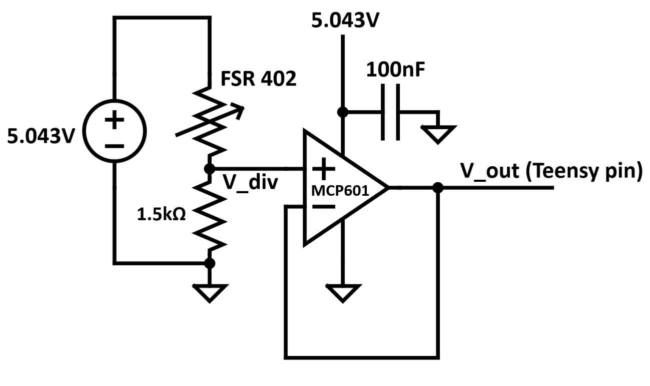


Fig. 17 Force sensor circuit schematic

The force sensor implements an FSR 402, which when a force is applied the effective resistance becomes lower. The FSR 402 has a rise time of less than 3 milliseconds. This is not an issue as significant changes in drag force should not happen in 3 milliseconds. The teensy sample rate of 10 Hz is deemed sufficient. The force sensor is placed in a voltage divider with a 1.5 kΩ resistor and powered with 5.043V (Figure 17).

$$V = \frac{1.5\ k\Omega}{1.5\ k\Omega + Rf} \cdot Vin \quad (8)$$

Rf represents the resistance of the force sensor and Vin is the input voltage of 5.043V as shown in equation 8. As the force increases the output voltage increases. The voltage divider is attached to a unity gain buffer using an MCP601 op-amp to prevent any loading from the power source. Based on the maximum thrust produced by the three motors the maximum speed of the robot was estimated to be 1 m/s but in practice, it was 0.5 m/s or less.

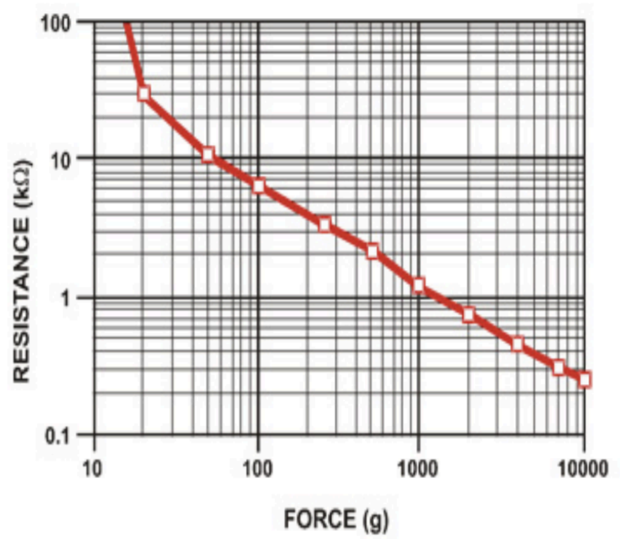


Fig. 18 Calibration Curve for Force Sensor [11]

Using the estimated maximum speed and COMSOL, the maximum effective drag force was calculated for each nose cone. The maximum drag force experienced by any of the nose cones was 7.4N. Using the calibration curve from the data sheet (Figure 18) of the FSR 402 the effective resistance was found to be around 1.5kΩ. Using the voltage divider Equation 8 the expected output voltage was 2.5V. The theoretical minimum voltage output is 0V when the robot is experiencing zero drag force. The circuit was designed to handle a drag force of 9.8N and output a maximum of 3.3V.

In practice with the housing attached, the rest voltage is 1.8V due to the pressure of the housing and water pressure. However, the sensor will still be in range. Using the calibration curve from the data sheet the force the sensor experiences from the housing and water pressure is 3.92N. The maximum force the sensor can experience before railing out is 9.8N. To experience this force the drag force needs to be 5.8N. Using the drag coefficient of a flat plate [12] and equation 1, the robot would need to go at a speed of 0.8m/s but the robot goes about 0.5m/s therefore the sensor will work with the added pressure.

The force sensor is calibrated by the voltage the teensy reads which is converted into the resistance of the force sensor using the voltage divider Equation 8. From this resistance using the calibration curve (Figure 18) the total force on the sensor is found. Subtracting 3.92N, the force of the housing and water pressure on the sensor, the drag force is found in newtons.

## V. Procedure and Data Processing

Using code adapted from the E80 GitHub repository, the robot was programmed to measure 5 trials per deployment by running the motors for 30 seconds at 30-second intervals [13]. A 180-second delay was set after starting the code to allow for disconnecting the teensy, closing the box, securing the box with the straps, attaching the nose cone, and pointing the robot away from the dock. The robot was followed in a kayak to ensure it was on course for each trial and to recover the robot after the 5 trials were done. The teensy measures in units representing 3.25mV, and all data was converted to voltage to be ready for processing. Processing was performed on the raw data to convert it to the desired quantities in Equation 1 which were then used to solve for the coefficient of drag. The average of the 5 trials per nose cone was taken to obtain the resulting drag coefficients.

### A. Velocity Sensor

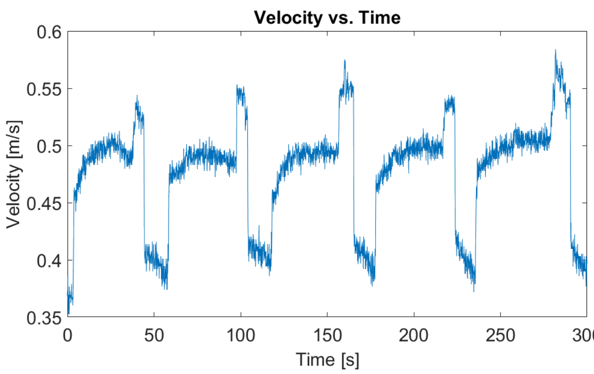

Fig. 19 Example of velocity data

The output voltages were converted to velocity using the calibration equations (Equation 4,5) and Equation 2. The trials were then narrowed to their respective 30-second window and averaged over to obtain the estimated terminal velocity. The trials were then averaged over to obtain the final velocity per nose cone. The sharp peaks at the end of each trial in Figure 19 indicate the robot being reset for this particular set of trials. This peak is not factored into the average as it does not represent the trial itself. It was not always the case that the robot was reset for every trial but this set of trials produced the best visual for velocity data.

### B. Anemometer

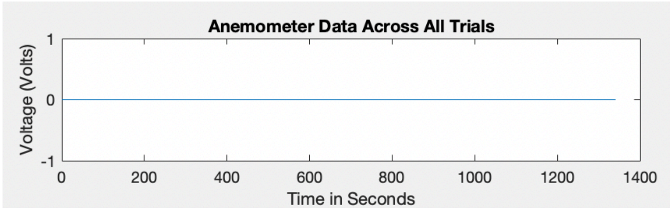

Fig. 20 Anemometer data across all trials

When deploying the robot at Dana Point the anemometer did not spin due to low wind conditions. This resulted in the teensy recording 0V across all trials (Figure 20). This is consistent with footage reviewed of deployment. There was one partial rotation, which likely did not result in the magnet aligning with the hall effect sensor as there are no spikes in the data.

### C. Weather Vane

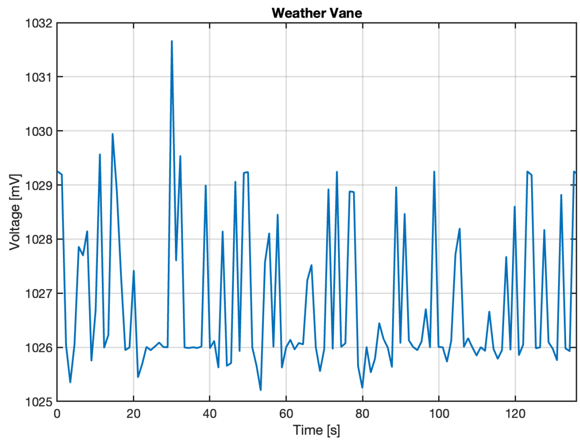

Fig. 21 Data collected from the weather vane

The data shows periods of inactivity from the weather vane (Figure 21). These periods, during which no voltage change is generated, indicate that the weather vane is not undergoing any rotation.

### D. Force Sensor

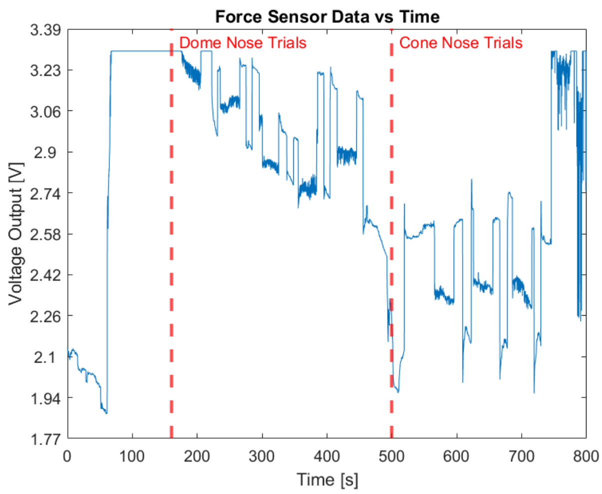


Fig. 22 Data from the force sensor

Force calculations only account for the time when the motors are turned on and the robot is moving with a forward velocity (Figure 22). The voltage output for each nose cone is determined by taking the average across all five trials. The voltage is converted using Equation 8 to get the resistance of the force sensor. There are only 3 trials for the cone-shaped nose cone since the force sensor became disconnected. A later test confirmed the force sensor was no longer working after deployment.

## VI. Deployment Safety

During the deployment of the robot, safety protocols were strictly adhered to in order to ensure both environmental and personal safety. Upon the time of deployment, a kayak, manned by two team members donned in life jackets, maintained close proximity to the robot. These precautions were taken to ensure that the robot did not deviate into unchartered areas, thereby maintaining control over its trajectory at all times.

Prior to each deployment, the team conducted a thorough cleaning of the housing unit. This step was crucial to ensure that the cable was reinserted into a clean environment, reducing the risk of any damage to the robot's components.

## VII. Results

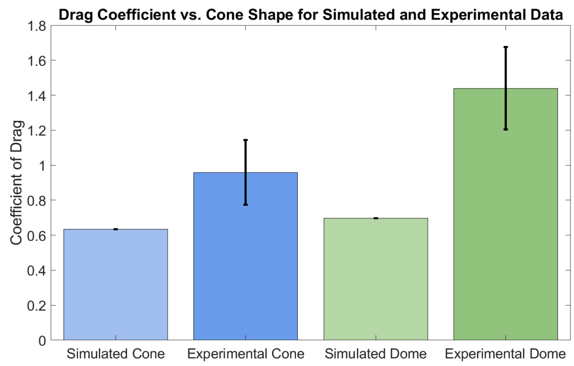


Fig. 23 Graph showing the coefficients of drag in the experiment and simulation for the cone and dome shaped noses.

The experimental velocities are 0.5308m/s for the cone-shaped nose and 0.4720m/s for the dome-shaped nose. The experimental drag forces are 2.45N and 2.9N for the dome-shaped nose. Wind does not need to be accounted for because the results indicated the wind speed was insignificant. Experimental force and velocity values are input to Equation 1 to obtain the drag coefficients of the noses. These are 0.959±0.185 and 1.44±0.235 respectively (Figure 23). Comparatively, the drag coefficients from the simulation are $0.636\pm10^{-4}$ and $0.696\pm10^{-4}$ respectively. Using the coefficients of drag from simulation and the experimental force values with Equation 1, the expected velocities of the cone-shaped and dome-shaped noses are 0.664m/s and 0.678m/s respectively. These values correspond to a 20% and 30% respective performance reduction between theoretical and experimental velocity.

Error propagation for the experimental coefficients of drag was calculated using the squared sum errors of drag, reference area, and velocity. The error in drag and velocity were obtained using their governing equations and the uncertainty in reference area radius was taken to be the standard measurement error $\frac{1}{\sqrt{12}}$mm. COMSOL determined the uncertainty in the simulation to be on the order of $10^{-4}$.

## VIII. Conclusions

The experimental results do not match the expected results from the simulation within the uncertainty bounds, however, they are consistent with the dome having more drag than the cone. The data suggests that the cone shape is more hydrodynamic than the dome because it has a smaller coefficient of drag. Additionally, the data suggests that field testing results in slower velocity than the simulation predicted. The discrepancy between simulated and field velocities was greater for the dome-shaped nose than the cone-shaped nose.

It is no surprise that an idealized simulation will yield different results than experimental measurements. A primary reason that results vary is that the ocean is not an idealized system like the one that COMSOL runs. There is no guaranteed laminar flow in the ocean and laminar flow is specifically simulated on COMSOL. The objects being tested do not have uniform surface characteristics in the measured tests but do on COMSOL. Additional sources of disparity are the additional drag from the robot apparatus, currents in the water, vibrations coming from the motor, and the added drag from torque due to the motor thrust being above the nose cone.

It is important to consider both simulation and in situ testing as this experiment has proven that both obtain different values. Practical experiments are able to account for nonidealities in manufacturing and the intended environment of use. In practice, the cones are not accurate in uncontrolled environments as they are still different from the simulated values.

On launch day there were countless setbacks that had to be overcome, proving the importance of testing sensors and code early and often. This additionally proved the importance of having backup sensors.

The results of this experiment could be furthered to optimize a nose cone that reduces experimental and theoretical discrepancy. Additionally, it would be useful to study the cone shapes in a controlled environment such as the wind tunnel in addition to the ocean. This would allow for comparison using similitude with the real object under test. A more comprehensive experiment would test a greater variety of shapes to provide a better understanding of an optimal shape.

## Acknowledgment

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## References

[1] M. Saghafi and R. Lavimi, "Optimal design of nose and tail of an autonomous underwater vehicle hull to reduce drag force using numerical simulation," Proceedings of the Institution of Mechanical Engineers, Part M: Journal of Engineering for the Maritime Environment, vol. 234, no. 1, pp. 76–88, Jul. 2019, doi: https://doi.org/10.1177/1475090219863191. [Accessed April 25, 2024]

[2] T. Benson, "Effect of Size on Drag," www.grc.nasa.gov. https://www.grc.nasa.gov/www/k-12/VirtualAero/BottleRocket/airplane/sized.html. [Accessed April 23, 2024]

[3] T. Benson, "The Drag Coefficient," www.grc.nasa.gov. https://www.grc.nasa.gov/www/k-12/VirtualAero/BottleRocket/airplane/dragco.html. [Accessed April 27, 2024]

[4] NXP Semiconductors, "i.MX RT1060 Datasheet", i.MX RT1060, 04/2019 [Accessed April 29, 2024]

[5] T. Benson, "Pitot Tube," www.grc.nasa.gov. https://www.grc.nasa.gov/www/k-12/VirtualAero/BottleRocket/airplane/pitot.html. [Accessed April 28, 2024]

[6] E80 Teaching Team, "Reference Anemometer Procedure," https://docs.google.com/document/d/1rY2bEjX3RP1EuHBjaoB50aiPz61I9q4W86KAzeNQdSo/edit. [Accessed April 3, 2024]

[7] Electronic Tutorials, "Hall effect Sensor," https://www.electronics-tutorials.ws/electromagnetism/hall-effect.html [Accessed April 25th, 2024]

[8] Diodes Incorporated, "Ultra High Sensitivity Micropower OmniPolar Hall-Effect Switch", AH9246, 09/15 [Accessed March 15, 2024]

[9] E80 Teaching Team, "Reference WeatherVane," https://docs.google.com/document/d/1f8EBwjwC41yVn9NPL-A11Ml1rJ3tqBN0ptcQc6E8O4U/edit. [Accessed April 4, 2024]

[10] Mouser Electronics, "Integrated Silicon Pressure Sensor On-Chip Signal Conditioned, Temperature Compensated and Calibrated", MPX5700, 10/2012 [Accessed April 3, 2024]

[11] Interlink Electronics, "FSR 402 Data Sheet", 30-81794 [Accessed March 15, 2024]

[12] "Shape Effects on Drag," www.grc.nasa.gov. [Accessed April 28, 2024]

[13] "Team-35/Default_Robot.ino at main · lanilei/Team-35." Accessed: Apr. 30, 2024. [Online]. Available: https://github.com/lanilei/Team-35/blob/main/Default_Robot.ino.

Code Appendix 1: Modified portion of the Surface Activity Code

```
if (currentTime > 180000 && currentTime <210000) {
  motor_driver.drive(-240,-240,-240);
  } else if (currentTime > 240000 && currentTime <270000) {
  motor_driver.drive(-240,-240,-240);
  } else if (currentTime > 300000 && currentTime <330000) {
  motor_driver.drive(-240,-240,-240);
  } else if (currentTime > 360000 && currentTime <390000) {
  motor_driver.drive(-240,-240,-240);
  } else if (currentTime > 420000 && currentTime <450000) {
  motor_driver.drive(-240,-240,-240);
    } else if (currentTime > 570000 && currentTime <600000) {
```

```
    motor_driver.drive(-240,-240,-240);
  } else if (currentTime > 630000 && currentTime <660000) {
    motor_driver.drive(-240,-240,-240);
  } else if (currentTime > 690000 && currentTime <720000) {
    motor_driver.drive(-240,-240,-240);
  } else if (currentTime > 750000 && currentTime <780000) {
    motor_driver.drive(-240,-240,-240);
  } else if (currentTime > 810000 && currentTime <840000) {
    motor_driver.drive(-240,-240,-240);
  } else if (currentTime > 960000 && currentTime <990000) {
    motor_driver.drive(-240,-240,-240);
  } else if (currentTime > 1020000 && currentTime <1050000) {
    motor_driver.drive(-240,-240,-240);
  } else if (currentTime > 1080000 && currentTime <1110000) {
    motor_driver.drive(-240,-240,-240);
  } else if (currentTime > 1140000 && currentTime <1170000) {
    motor_driver.drive(-240,-240,-240);

  } else if (currentTime > 1200000 && currentTime <1230000) {
    motor_driver.drive(-240,-240,-240);
  } else {
    motor_driver.drive(0,0,0);
```